



Applied Machine Learning

COMP 551 001

April 22 2:00 pm – 5:00 pm

EXAMINER: Yue Li

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STUDENT NAME:		McGILL ID:										
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EXAM:	CLOSED BOOK ☐ OPEN BOOK ☒
	SINGLE-SIDED ☐ PRINTED ON BOTH SIDES OF THE PAGE ☐
	MULTIPLE CHOICE ANSWER SHEETS: YES ☒ NO ☐ NOTE: The Examination Security Monitor Program detects pairs of students with unusually similar answer patterns on multiple-choice exams. Data generated by this program can be used as admissible evidence, either to initiate or corroborate an investigation or a charge of cheating under Section 17 of the Code of Student Conduct and Disciplinary Procedures.
	ANSWER BOOKLET REQUIRED: YES ☐ NO ☒
	EXTRA BOOKLETS PERMITTED: YES ☐ NO ☒
	ANSWER ON EXAM: YES ☒ NO ☐
	SHOULD THE EXAM BE: RETURNED ☒ KEPT BY STUDENT ☐
CRIB SHEETS:	PERMITTED ☒ <u>Specifications:</u> (ex: one 8 1/2X11 handwritten double-sided sheet) NOT PERMITTED ☐
DICTIONARIES:	TRANSLATION ONLY ☒ REGULAR ☒ NOT PERMITTED ☐
CALCULATORS:	NOT PERMITTED ☐ PERMITTED (Non-Programmable) ☒ PERMITTED (Programmable) ☐
ANY SPECIAL INSTRUCTIONS: e.g. molecular models	

- You have 180 minutes to write the exam.
- Hard-copy notes, books, and printed slides are allowed but electronic devices are NOT allowed.
- This exam contains 24 questions on 18 pages for total of 100.
- 18 multiple-choice questions: answer each question on the SCANTRON sheet provided to you.
- 6 short-answer questions: write your answer directly below each question.
- Advice: Try not to spend too much time searching answers through your notes as it will slow you down and you will not have enough time to complete this exam in 3 hours. Good luck!

1 Multiple-choice questions (54 points)

1. (3 points) You trained a machine learning model to predict the protein content of different food with the mean squared error loss function. Your model has a hyperparameter λ and you need to decide which value of λ to use. You have the following table for the different losses as function of the value of lambda:

λ	training loss	validation loss	test loss
10	30	31	29
3	15	14	16
1	9	16	13
0.3	5	17	18
0.1	2	20	21

Which value of λ should you select

- A. 10
- B. 3**
- C. 1
- D. 0.3
- E. 0.1

Solution: $\lambda = 3$ minimized the validation loss.

2. (3 points) Looking at the table in question 1, for which value of λ does the model have the highest bias?

- A. 10**
- B. 3
- C. 1
- D. 0.3
- E. 0.1

Solution: $\lambda = 10$ has the highest training loss.

3. (3 points) In the context of bias-variance trade-off, which of the following actions will *most likely* decrease the model variance?
- A. Increase tree-depth in the regression tree.
 - B. Increase the number of attention heads in a Transformer model.
 - C. Increase K in K -nearest neighbours.**
 - D. Increase the number of channels in the hidden layer of a CNN.
 - E. Increase the width of an MLP layer.

Solution: Increasing K in K-NN increases bias and reduces variance.

4. (3 points) In the context of bias-variance trade-off, which of the following actions will *most likely* decrease the model bias?
- A. Use a smaller training set in K -nearest neighbours.
 - B. Reduce the size of the hidden state in an RNN.
 - C. Train for fewer epochs.
 - D. Increase the L1 regularization strength.
 - E. Increase the size of a CNN Kernel.**

Solution: Increasing Kernel size makes for a more expressive model, thus reducing bias.

5. (3 points) You are working for a big news agency and you want to build an LLM to automatically tag news stories with a variety of labels. Example tags include: politics, sports, canada, economy, lifestyle. Your set of tags is known in advance and each news story can potentially have multiple tags (for example both "canada" and "politics" for a news story about federal elections). What should the last (output) layer of your model be like?
- A. Single neuron, Sigmoid activation.
 - B. Multiple neurons, Softmax activation.
 - C. Multiple neurons, Sigmoid activation.**
 - D. Multiple neurons, Linear activation.
 - E. Multiple neurons, ReLU activation.

Solution: This is a multilabel regression problem.

6. (3 points) For a given threshold value, you obtain the following confusion matrix:

	Predicted Positive	Predicted Negative	Total
Positive	16	4	20
Negative	6	14	20
Total	22	18	40

Where would you put the point for this threshold on the ROC curve (to the nearest decimal point)?

- A. (0.2, 0.4)
- B. (0.2, 0.7)
- C. (0.2, 0.8)
- D. (0.3, 0.8)**
- E. (0.3, 0.7)

Solution: The true positive rate is $16/20=0.8$ The false positive rate is $6/20 = 0.3$

7. (3 points) We are building a predictive model for an architecture firm which needs to predict the cost (in dollars) of architectural project proposals for which the firm competes. Which of the following losses should you use to train your model? Let $\hat{y}$ be the prediction of your model and y be the true cost of the project.
- A. $(y - \hat{y})^2$**
 - B. $-y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$
 - C. $-\sum_c y_c \log(\hat{y}_c)$
 - D. $-\sum_c (y_c \log(\hat{y}_c) + (1 - y_c) \log(1 - \hat{y}_c))$
 - E. $\max(0, 1 - y\hat{y})$

Solution: This is a regression problem, for a continuous value.

8. (3 points) We are training your neural network with SGD and you quickly realize that your gradients are too noisy and you need smoother gradients as a function of epochs. Which of the following will NOT make your gradients smoother?
- A. Adding or increasing momentum.
 - B. Using the sum of squared errors rather than mean squared errors as a loss function.**
 - C. Increasing the minibatch size.
 - D. Lowering the learning rate.
 - E. Using Gradient Descent instead SGD.

Solution: Using sum of squared errors rather mean squared errors will multiply your loss function by the batch size, effectively multiplying your learning rate by the batch size which means you will be taking larger gradient steps and thus your gradients will vary more from one step to the next.

9. (3 points) We are training an MLP with SGD. Your MLP is deep to the level that you are suffering from a vanishing gradient problem. Which of the following activation functions could you use in the hidden layers to help alleviate the vanishing gradient problem?
- A. $\tanh(x) = \frac{\exp(2x)-1}{\exp(2x)+1}$
 - B. $\sigma(x) = \frac{1}{1+\exp(-x)}$
 - C. Clipped ReLU: $\text{CReLU}(x) = \min\{1, \max\{0, x\}\}$.
 - D. $\log(1 + \exp(a))$**

E. Sigmoid with a steeper derivative: $\sigma(x) = \frac{1}{1+\exp(-100x)}$

Solution: Only

$$\log(1 + \exp(a))$$

(Softplus) does not have a gradient which goes to zero at the extremities (i.e., saturation point).

10. (3 points) We have a training dataset with 1000 features but only 50 training examples on which you are trying to fit a linear regression. Your validation error is extremely large. What could you do to lower the validation error.

- A. Add more non-linear basis functions.
- B. Train using Gradient Descent rather than finding the solution analytically.
- C. Lower the learning rate in SGD.
- D. Make sure that the dataset is linearly separable.
- E. Add an L2 regularization term.**

Solution: You are likely suffering from overfitting, and thus need to reduce variance.

11. (3 points) Which of the following regularization can lead to a solution, where either all D weights are exactly zeros or none of the D weight is exactly 0?

- A. $\lambda \|\mathbf{w}\|_2^2 = \lambda \sum_{d=1}^D w_d^2$
- B. $\lambda \|\mathbf{w}\|_2 = \lambda \sqrt{\sum_{d=1}^D w_d^2}$**
- C. $\lambda \|\mathbf{w}\|_1 = \lambda \sum_{d=1}^D |w_d|$
- D. $\lambda_1 \|\mathbf{w}\|_2^2 + \lambda_2 \|\mathbf{w}\|_1$
- E. $\lambda \sum_{g=1}^G \|\mathbf{w}_g\|_2$, where $\|\mathbf{w}_g\|_2 = \sqrt{\sum_{d \in \mathcal{S}_g} w_d^2}$ and $\mathcal{S}_g \subset \{1, \dots, D\}$

Solution: $\lambda \|\mathbf{w}\|_2$ is a special case of group lasso where all features form one group. The gradient is undefined when $\sum_d w_d^2 = 0$:

$$\frac{\partial \sqrt{\sum_d w_d^2}}{\partial w_d} = \frac{\partial}{\partial w_d} (\sum_d w_d^2)^{1/2} = \frac{w_d}{\sqrt{\sum_d w_d^2}} \quad (\text{undefined when } \sum_d w_d^2 = 0)$$

12. (3 points) Suppose we want to apply $L1$ -regularization with weight penalty λ to the weights in an MLP for regression by minimizing sum of squared error loss. We apply non-linear activation functions for all hidden units. Which of the layer weights can we directly optimize using the *coordinate descent* algorithm that we learn in training the LASSO regression?

- A. input-to-hidden weights
- B. hidden-to-hidden weights
- C. hidden-to-output weights for regression**

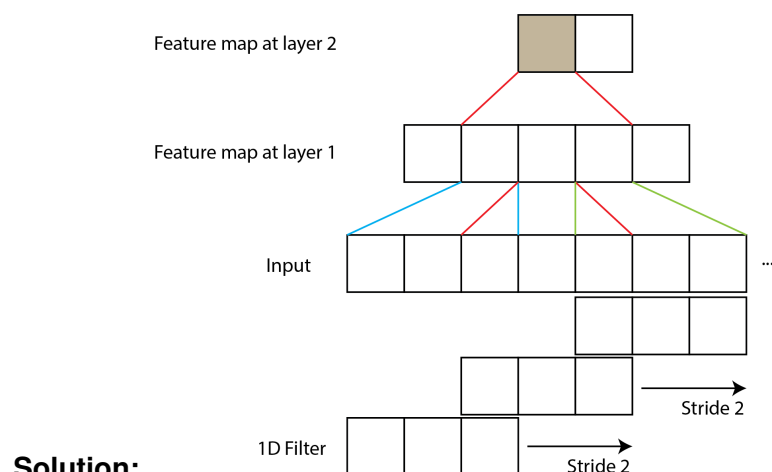
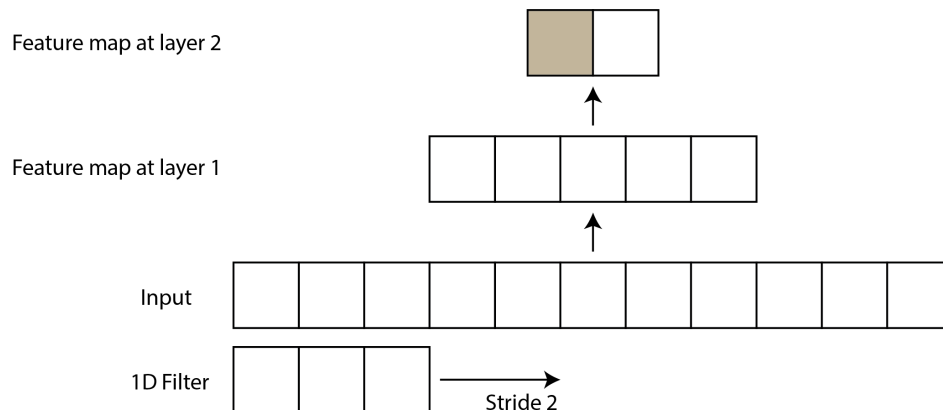
- D. We cannot apply coordinate descent to any of the layers
 E. We can apply coordinate descent to *all* weights of the layers

Solution: Coordinate descent works on hidden-to-output as we can decouple feature from the rest of the features and update the $\hat{w}_k^{\{o\}}$ while fixing the rest of the output weights:

$$\hat{w}_k^{\{o\}} = \begin{cases} \frac{\mathbf{z}_k^\top (\mathbf{y} - \hat{\mathbf{y}}_{\setminus k}) + \lambda}{\|\mathbf{z}_k\|_2^2} & \mathbf{z}_k^\top (\mathbf{y} - \hat{\mathbf{y}}_{\setminus k}) < -\lambda \\ 0 & \mathbf{z}_k^\top (\mathbf{y} - \hat{\mathbf{y}}_{\setminus k}) \in [-\lambda, \lambda] \\ \frac{\mathbf{z}_k^\top (\mathbf{y} - \hat{\mathbf{y}}_{\setminus k}) - \lambda}{\|\mathbf{z}_k\|_2^2} & \mathbf{z}_k^\top (\mathbf{y} - \hat{\mathbf{y}}_{\setminus k}) > \lambda \end{cases}$$

13. (3 points) We build a 1D CNN, where each convolution layer has kernel size of 3 and a stride of 2. What is the receptive field of each pixel in the 1D output feature map from convolution layer 2 (i.e., the shaded cell in the figure below) with respect to the 1D input?

- A. 3
 B. 4
 C. 5
D. 7
 E. 9



Solution:

14. (3 points) We are using bagging to train an ensemble of models. The training set has 3 samples. Each bagging model will be trained on a different training set. How many possible training sets can models in the ensemble be trained on?
- A. 1
 - B. 27
 - C. 12
 - D. 5
 - E. 10**

Solution: As shown in the final review session, the number of distinct bootstrap samples can be calculated as $\binom{2N-1}{N-1}$. You may also manually enumerate 10 distinct sets as 111, 222, 333, 112, 113, 221, 223, 331, 332, 123.

For this question, there was an error in the original printout. We provided a separate page for this question. we first check the answers on the Scantron. If it is incorrect, then we checked their answer manually on the exam sheet.

15. (3 points) Which of the following statement about boosting is **false**?
- A. Boosting cannot train weak learners in parallel.
 - B. Boosting can help reduce model bias.
 - C. Boosting can work with both regression and classification problems.
 - D. The m^{th} weak learner is trained to reduce the error made by the ensemble model that uses $m - 1$ weak learners.
 - E. Boosting will always work well with any supervised machine learning models by treating them as weak learners.**

Solution: Boosting sequentially trains models to learn the remaining errors, it can thus reduce the bias of the original model. However, boosting relies on combining simple models. It does not work well with highly complex models, which is prone to overfitting. It will not work well with multiple linear models either because of the additive effects of the first multiple linear regression model will make adding another multiple linear regression model completely irrelevant.

16. (3 points) You train an RNN to predict agricultural crop yield (a single number which is determined after the harvest, i.e., seq2vec) by passing the last hidden state of the RNN through a fully connected output layer with linear activation. The RNN has as input the daily weather patterns of the last 60 days before harvest. You get good results but decide to retrain the same model on 90 days of daily weather pattern before harvest instead, in the hope of getting better results. Which of the following statements is **false**?
- A. The initialization of the hidden state will have less of an effect.
 - B. The model will use the same amount of memory at test time.
 - C. The model will backpropagate more gradients.
 - D. The model will have more trainable parameters.**
 - E. The vanishing gradient problem becomes more prominent in training RNN on 90-day sequences than in training RNN on 60-day sequences.

Solution: RNN use weight-sharing in time so the model does not have more trainable parameters. The model will use more memory during training because the backpropagation graph will be bigger and take up more memory but will not use more memory at inference time because the hidden state is of constant size.

17. (3 points) What is the loss term that should **not** be in the *full* loss function (including the orthogonal and orthonormal constraints) for deriving the third principal component (PC) $\mathbf{w}_3$ given the first and the second PC? The λ 's below are the Lagrange multipliers.

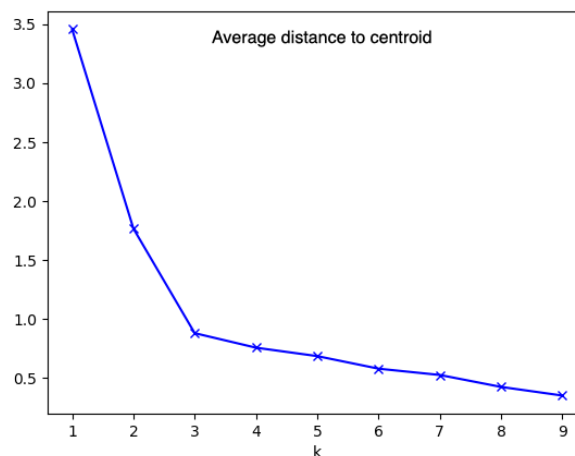
- A. $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - z_{1,n}\hat{\mathbf{w}}_1 - z_{2,n}\hat{\mathbf{w}}_2 - z_{3,n}\mathbf{w}_3\|_2^2$
- B. $\lambda_3(\mathbf{w}_3^\top \mathbf{w}_3 - 1)$
- C. $\lambda_{23}(\mathbf{w}_2^\top \mathbf{w}_3)$
- D. $\lambda_{13}(\mathbf{w}_1^\top \mathbf{w}_3)$
- E. $\lambda_4 \|\mathbf{w}_3\|_2^2$

Solution:

$$\mathcal{L}(\mathbf{w}_3) = \frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - z_{1,n}\hat{\mathbf{w}}_1 - z_{2,n}\hat{\mathbf{w}}_2 - z_{3,n}\mathbf{w}_3\|_2^2 + \lambda_3(\mathbf{w}_3^\top \mathbf{w}_3 - 1) + \lambda_{23}(\mathbf{w}_2^\top \mathbf{w}_3) + \lambda_{13}(\mathbf{w}_1^\top \mathbf{w}_3)$$

Standard PCA does not have the L_2 -regularization constraint.

18. (3 points) You are using K-means clustering. You want to use the elbow method to decide on which is the best k to use so you plot the average distance of points to their centroid as a function of k to get the following graph. Which k should you choose?



- A. 1
- B. 3**
- C. 5
- D. 7
- E. 9

Solution: The kink (i.e., inflection point) in the slope happens for $k=3$.

2 Short answer questions (46 points)

Partial marks were given to the answers that are outside of the rubrics.

19. (5 points) Suppose we design a 1-layer MLP with K hidden units and 1 output for regression. The forward pass for one training data point $\mathbf{x} \in \mathbb{R}^{1 \times D}$ is:

$$\begin{aligned}a_k &= \mathbf{x} \mathbf{w}_k^{\{1\}} \\z_k &= \frac{\exp(a_k)}{\sum_{k=1}^K \exp(a_k)} \\\hat{y} &= \mathbf{z} \mathbf{w}^{\{o\}} \\L &= (\hat{y} - y)^2\end{aligned}$$

Write down the exact mathematical expression of the partial derivative for input-to-hidden weights $w_{d,k}$ for input feature $d \in \{1, \dots, D\}$ and hidden unit $k \in \{1, \dots, K\}$ by the chain rule.

Hint: The activation in the hidden layer is a softmax transformation of the pre-activated hidden activities a_k 's over all K hidden units (whereas the standard MLPs have activation applied to each individual hidden unit, e.g., $z_k = \phi(a_k)$). Therefore, there is a dependency among hidden units due to the softmax transformation – similar to the dependency among the outputs in multi-class regression.

Solution: Accepted answer 1:

$$\frac{\partial L}{\partial w_{d,k}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z_k} \sum_{j=1}^K \frac{\partial z_j}{\partial a_k} \frac{\partial a_k}{\partial w_{d,k}}$$

Accepted answer 2 (backprop):

$$\begin{aligned}\frac{\partial L}{\partial \hat{y}} &= 2(\hat{y} - y) \\ \frac{\partial L}{\partial z_k} &= \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z_k} \\ \frac{\partial L}{\partial a_k} &= \sum_j \frac{\partial L}{\partial z_j} \frac{\partial z_j}{\partial a_k} \\ \frac{\partial L}{\partial w_{d,k}} &= \frac{\partial L}{\partial a_k} \frac{\partial a_k}{\partial w_{d,k}}\end{aligned}$$

Here it's OK if they leave the answer like the above without expanding the last partial.

Also, it is OK they wrote down the exact form of the partial for say $\frac{\partial z_j}{\partial a_k}$ but we don't need to deduct point if they got it wrong as long as they got the above partial chain-rule expression right.

Deduct 2 point, if the answer is something like this:

$$\frac{\partial L}{\partial w_{d,k}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z_k} \sum_{j=1}^K \frac{\partial z_k}{\partial a_j} \frac{\partial a_j}{\partial w_{d,k}}$$

Deduct 4 points, if the answer is:

$$\frac{\partial L}{\partial w_{d,k}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z_k} \frac{\partial z_k}{\partial a_k} \frac{\partial a_k}{\partial w_{d,k}}$$

20. (10 points) Consider the following input $\mathbf{X}$ and convolution filter $\mathbf{W}$.

- (8 pts) [original question] When using zero padding of 1 and stride of 2, what is the output $\mathbf{A}$?
- (8 pts) [updated question] With no zero padding and stride of 1, what is the output $\mathbf{A}$?
- (2 pts) After the convolution layer, a ReLU activation non-linearity is applied followed by a 2×2 max-pooling (i.e., $\text{maxpool}_{2 \times 2}(\text{ReLU}(\mathbf{A}))$). What is the resulting output?

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 2 \\ 4 & 1 & 1 & 2 \\ 6 & 1 & 3 & 2 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution:

[original version]

Assuming the filter stops at the last zero-padded column or row after taking a stride of 2 once.

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix} \text{ and } h = [3]$$

[updated version]

$$\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 9 & 0 \end{bmatrix} \text{ and } h = [9].$$

21. (10 points) Suppose we are using a two-headed self-attention transformer module to compute the embeddings of two time points. Input at time 1 is $\mathbf{x}_1 = [0, 1, 0]$ and input at time 2 is $\mathbf{x}_2 = [0, 0, 1]$. Each head $i \in \{1, 2\}$ computes the embedding as:

$$\mathbf{A}_i = \text{softmax}(\mathbf{Q}_i \mathbf{K}_i^\top), \quad \mathbf{Z}_i = \mathbf{A}_i \mathbf{V}_i$$

where $\mathbf{q}_{t,i} = \mathbf{x}_t \mathbf{W}_i^{(query)}$, $\mathbf{k}_{t,i} = \mathbf{x}_t \mathbf{W}_i^{(key)}$, $\mathbf{a}_{t,i} = \text{softmax}(\mathbf{q}_{t,i} \mathbf{K}_i^\top)$ and $\mathbf{v}_{t,i} = \mathbf{x}_t \mathbf{W}_i^{(value)}$; therefore by concatenating rows (i.e., time points), we have $\mathbf{Q}_i = [\mathbf{q}_{1,i}; \mathbf{q}_{2,i}]$, $\mathbf{K}_i = [\mathbf{k}_{1,i}; \mathbf{k}_{2,i}]$, $\mathbf{A}_i = [\mathbf{a}_{1,i}; \mathbf{a}_{2,i}]$, and $\mathbf{V}_i = [\mathbf{v}_{1,i}; \mathbf{v}_{2,i}]$.

For the ease of computation, in this question, we ignore the inverse scaling $\sqrt{D}$ so we are NOT using $\mathbf{A} = f(\mathbf{Q}_i \mathbf{K}_i^\top / \sqrt{D})$ but rather $\mathbf{A} = f(\mathbf{Q}_i \mathbf{K}_i^\top)$ for in this question.

Using the following weights:

$$\begin{aligned} \mathbf{W}_1^{(query)} &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{W}_2^{(query)} = \begin{bmatrix} 0 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{W}_1^{(key)} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{W}_2^{(key)} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \\ \mathbf{W}_1^{(value)} &= \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{W}_2^{(value)} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

1. (7 pts) Compute the output embedding for either head 1 or head 2 for time step 2 (i.e., to save your time, you don't need to compute embedding for both heads)? There is no need to calculate exponential and you can write your answer in terms of the exponential. Show your work.
2. (3 pts) What's the dimension of the output embedding for time 2 if we concatenate the embedding from the two heads? Specify both the number of rows and the number of columns in your answer.

Solution:

1.

$$\mathbf{q}_1 = \mathbf{x}_2 \mathbf{W}_1^{(query)} = [0, 1] \quad (1)$$

$$\mathbf{q}_2 = \mathbf{x}_2 \mathbf{W}_2^{(query)} = [1, 0] \quad (2)$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{K}_1 = \mathbf{X} \mathbf{W}_1^{(key)} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \quad (3)$$

$$\mathbf{K}_2 = \mathbf{X} \mathbf{W}_2^{(key)} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \quad (4)$$

$$\mathbf{V}_1 = \mathbf{X} \mathbf{W}_1^{(value)} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad (5)$$

$$\mathbf{V}_2 = \mathbf{X} \mathbf{W}_2^{(value)} = \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \quad (6)$$

$$\mathbf{q}_1 \mathbf{K}_1^\top = [1, 0] \quad (7)$$

$$\mathbf{q}_2 \mathbf{K}_2^\top = [0, 1] \quad (8)$$

$$\mathbf{a}_1 = \text{Softmax}(\mathbf{q}_1 \mathbf{K}_1^\top) = \frac{1}{1+e} [e, 1] = \left[\frac{e}{1+e}, \frac{1}{1+e} \right] \quad (9)$$

$$\mathbf{a}_2 = \text{Softmax}(\mathbf{q}_1 \mathbf{K}_1^\top) = \frac{1}{1+e} [1, e] = \left[\frac{1}{1+e}, \frac{e}{1+e} \right] \quad (10)$$

$$\mathbf{h}_1 = \mathbf{a}_1 \mathbf{V}_1 = \begin{bmatrix} \frac{e}{1+e} & \frac{1}{1+e} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{e}{1+e} & \frac{1-e}{1+e} \end{bmatrix} \quad (11)$$

$$\mathbf{h}_2 = \mathbf{a}_2 \mathbf{V}_2 = \begin{bmatrix} \frac{1}{1+e} & \frac{e}{1+e} \end{bmatrix} \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{-e}{1+e} & \frac{e}{1+e} \end{bmatrix} \quad (12)$$

$$\mathbf{z} = [\mathbf{h}_1, \mathbf{h}_2] = \frac{1}{1+e} [e, 1-e, -e, e] \quad (13)$$

2. Output dimension for time point 2 is 1×4 as shown in Equation (13)

For part 1:

- 3 pts for correct calculation of $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$.
- 2 pts for correct calculation of the softmax attention
- 2 pts for correct calculation of the embedding $\mathbf{z}_2$

22. (10 points) We have the following dataset of 4 (2D) points:

$$\mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & -2 \\ -1 & 1 \\ -2 & 1 \end{bmatrix}$$

We wish to cluster the points using K-Means with $K = 2$, i.e., two clusters. We initialize our clusters with $\mathbf{c}_1 = [1, -1]$ and $\mathbf{c}_2 = [-1, 1]$.

- (2 pts) Which points belongs to which cluster initially?
- (6 pts) After the first update of the centroids, what are the new values of the centroids?
- (2 pts) What do the centroids converge to and which points belong to which cluster?

Solution: The first two points belong to the first cluster. The last two points to the second cluster. The updated value for the centroid of the first cluster is thus the average of the first two points thus $c_1 = [1, -1.5]$ and the updated value for the centroid of the second cluster is thus the average of the last two points thus $c_2 = [-1.5, 1]$.

After the update of the centroids, the first two points still belong to the first cluster and the second two points still belong to the second cluster, thus the algorithm has converged.

23. (8 points) You have an input sequence $\mathbf{x}_0 = [1, 0]$, $\mathbf{x}_1 = [0, 1]$. You are using a vanilla RNN such that $h_{t+1} = \text{ReLU}(\mathbf{h}_t \mathbf{W}_h + \mathbf{x}_t \mathbf{W}_x + b)$. If

$$\mathbf{W}_x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{W}_h = \begin{bmatrix} 0 & 2 \\ 0.5 & 0 \end{bmatrix}, \quad \mathbf{b} = [0, 0], \quad \mathbf{h}_0 = [0, 0]$$

- (6 pts) What is $\mathbf{h}_2$ equal to? Show your work.
- (2 pts) In seq-to-seq aligned prediction, the RNN outputs $y_t = \mathbf{h}_t \mathbf{w}_y$, with $\mathbf{w}_y = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$, what is the output sequence (y_1, y_2) by this RNN on the input sequence (x_0, x_1) defined above?

Solution:

$$h_1 = \text{ReLU}(0 + [1, 0] + 0) = [1, 0] \quad (14)$$

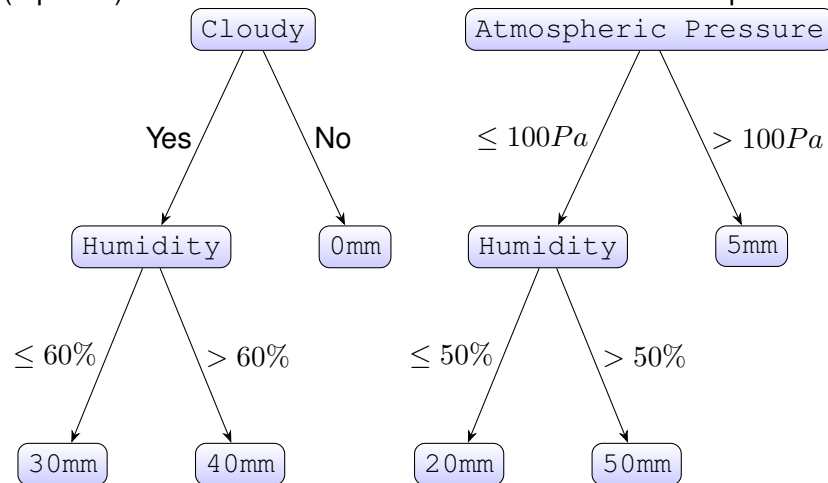
$$h_2 = \text{ReLU}([1, 0] \begin{bmatrix} 0 & 2 \\ 0.5 & 0 \end{bmatrix} + [0, 1]) = [0, 3] \quad (15)$$

$$y_1 = [1, 0] \begin{bmatrix} 2 \\ 0.5 \end{bmatrix} = 2 \quad (16)$$

$$y_2 = [0, 3] \begin{bmatrix} 2 \\ 0.5 \end{bmatrix} = 1.5 \quad (17)$$

So $(y_1, y_2) = (2, 1.5)$

24. (3 points) You trained a random forest of two trees to predict how much rain will fall in milliliters.



The first tree has a weight of 0.7 and the second tree has a weight of 0.3. If it is cloudy with a humidity level of 55% and an atmospheric pressure of 100 Pa, how many mm of rain do you predict?

Solution: $0.7 * 30\text{mm} + 0.3 * 50\text{mm} = 21\text{mm} + 15\text{mm} = 36\text{mm}$ of rain.

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